RISC-V AI Accelerator: End-to-End Run Process and Testing

1. Simulation & Software Build

- **RTL & Firmware Tests:** All simulation testbenches, firmware builds, and INT8 MAC verification checks passed completely.

```
ronok@rk ~/Documents/riscv-resilient-ai-accelerator % ./scripts/run_tests.sh
[BUILD] Compiling RV32I firmware
[BUILD] Generating firmware binary
[BUILD] Generating 32-bit HEX image
[BUILD] Generated 187 words
[PASS] Firmware build completed
==================================
[TEST] picorv32_baseline
==================================
MMIO WRITE: value=12
PASS: PicoRV32 executed RV32I instructions correctly.
simulation/testbench/tb_picorv32.v:90: $finish called at 315000 (1ps)
[PASS] picorv32_baseline
==================================
[TEST] picorv32_firmware
==================================
WARNING: simulation/testbench/tb_picorv32_firmware.v:68:
$readmemh(firmware/build/firmware.hex): Not enough words in the file for the requested range
[0:4095].
==================================
PicoRV32 Full Firmware & UART Test
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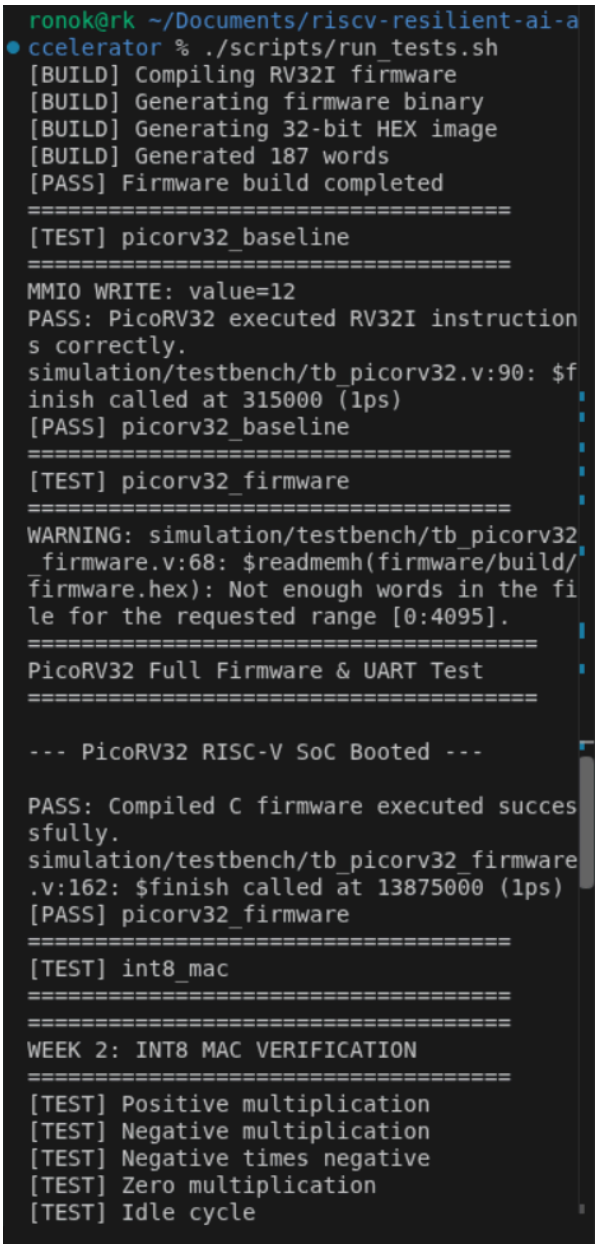
--- PicoRV32 RISC-V SoC Booted ---

PASS: Compiled C firmware executed successfully.
simulation/testbench/tb_picorv32_firmware.v:162: $finish called at 13875000 (1ps)
[PASS] picorv32_firmware
==================================
[TEST] int8_mac
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==================================
WEEK 2: INT8 MAC VERIFICATION
==================================
```

```
[TEST] Positive multiplication
[TEST] Negative multiplication
[TEST] Negative times negative
[TEST] Zero multiplication
[TEST] Idle cycle
[TEST] Clear accumulator
[TEST] Clear priority over MAC
[TEST] Signed INT8 boundaries
[TEST] Randomized verification
[TEST] 32-bit overflow wraparound
====================================
[PASS] All INT8 MAC tests passed
Total checks: 213
====================================
simulation/testbench/tb_int8_mac.v:163: $finish called at 2156000 (1ps)
[PASS] int8_mac
====================================
[TEST] int8_mac_exhaustive
====================================
========================================
INT8 MAC EXHAUSTIVE VERIFICATION
========================================
========================================
[PASS] Exhaustive INT8 MAC verification
Operand combinations checked: 65536
========================================
simulation/testbench/tb_int8_mac_exhaustive.v:89: $finish called at 655390000 (1ps)
[PASS] int8_mac_exhaustive
====================================
[TEST] int8_mac_fault
====================================
========================================
WEEK 3: FAULT INJECTION & RESILIENCE TEST
========================================
[TEST] Single-bit flip sweep on weights (0 to 7)
[TEST] Clamping mitigation on MSB bit-flip
========================================
[PASS] All Fault Resilience checks passed
Checks completed: 9
========================================
simulation/testbench/tb_int8_mac_fault.v:131: $finish called at 206000 (1ps)
[PASS] int8_mac_fault
====================================
[TEST] int8_mac_mmio
```

```
==================================
========================================
WEEK 2: INT8 MAC MMIO VERIFICATION
========================================
[TEST] Reset state
[TEST] Positive signed MAC
[TEST] Sticky DONE status
[TEST] Negative operand accumulation
[TEST] Negative times negative
[TEST] Partial writes ignored
[TEST] Clear accumulator
[TEST] Reserved register read
[TEST] Held mem_valid handshake
========================================
[PASS] All MMIO verification tests passed
Checks completed: 16
========================================
simulation/testbench/tb_int8_mac_mmio.v:273: $finish called at 1050000 (1ps)
[PASS] int8_mac_mmio
==================================
[TEST] picorv32_accelerator
==================================
========================================
WEEK 2: RISC-V ACCELERATOR FIRMWARE BUILD
========================================
[BUILD] Compiling RV32I accelerator firmware
[BUILD] Generating firmware binary
[BUILD] Generating 32-bit HEX image
[BUILD] Generated 127 words
[PASS] Accelerator firmware build completed
========================================
rtl/cpu/picorv32.v:1352: warning: @* is sensitive to all 32 words in array 'cpuregs'.
rtl/cpu/picorv32.v:1353: warning: @* is sensitive to all 32 words in array 'cpuregs'.
WARNING: simulation/testbench/tb_picorv32_accelerator.v:113:
$readmemh(firmware/accelerator/build/accelerator.hex): Not enough words in the file for the
requested range [0:4095].
========================================
WEEK 2: PICO RV32 + INT8 MAC SOC TEST
========================================
[INFO] RV32I firmware loaded
[INFO] MAC MMIO base: 0x80000000
[MAC] START command 1
[MAC] START command 2
[MAC] START command 3
```

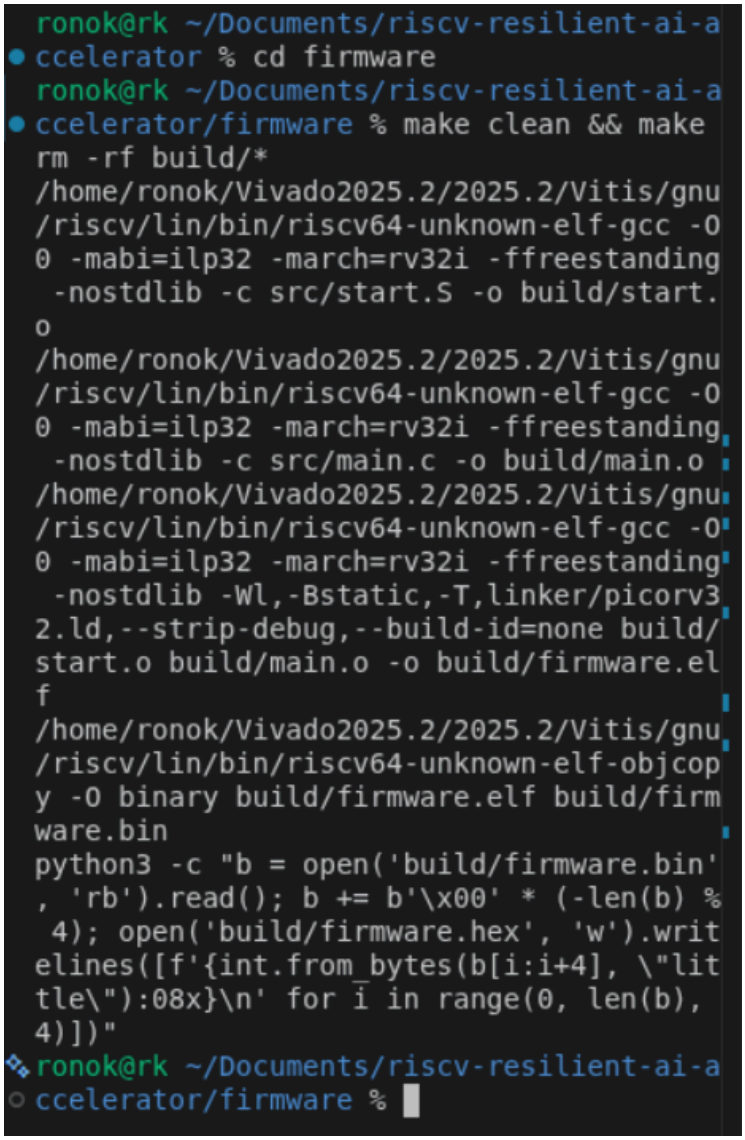
[RESULT] Firmware reported: 39
==========
[PASS] PicoRV32 controlled INT8 MAC
[PASS] Three MAC commands completed
[PASS] Final accumulator = 39
==========
simulation/testbench/tb_picorv32_accelerator.v:207: $finish called at 4325000 (1ps)
[PASS] picorv32_accelerator
==========
[SUCCESS] All RTL and firmware tests passed
==========
ronok@rk ~/Documents/riscv-resilient-ai-a



- **Firmware Compilation:** Bare-metal C code compiled successfully using the Vitis RISC-V GCC toolchain into a 32-bit little-endian HEX image.

Output:

```
ronok@rk ~/Documents/riscv-resilient-ai-accelerator/firmware % make clean && make
rm -rf build/*
/home/ronok/Vivado2025.2/2025.2/Vitis/gnu/riscv/lin/bin/riscv64-unknown-elf-gcc -O0
-mabi=ilp32 -march=rv32i -ffreestanding -nostdlib -c src/start.S -o build/start.o
/home/ronok/Vivado2025.2/2025.2/Vitis/gnu/riscv/lin/bin/riscv64-unknown-elf-gcc -O0
-mabi=ilp32 -march=rv32i -ffreestanding -nostdlib -c src/main.c -o build/main.o
/home/ronok/Vivado2025.2/2025.2/Vitis/gnu/riscv/lin/bin/riscv64-unknown-elf-gcc -O0
-mabi=ilp32 -march=rv32i -ffreestanding -nostdlib
-Wl,-Bstatic,-T,linker/picorv32.ld,--strip-debug,--build-id=none build/start.o build/main.o -o
build/firmware.elf
/home/ronok/Vivado2025.2/2025.2/Vitis/gnu/riscv/lin/bin/riscv64-unknown-elf-objcopy -O binary
build/firmware.elf build/firmware.bin
python3 -c "b = open('build/firmware.bin', 'rb').read(); b += b'\x00' * (-len(b) % 4);
open('build/firmware.hex', 'w').writelines([f'{int.from_bytes(b[i:i+4], \"little\"):08x}\n' for i in
range(0, len(b),4)])"
ronok@rk ~/Documents/riscv-resilient-ai-accelerator/firmware %
```



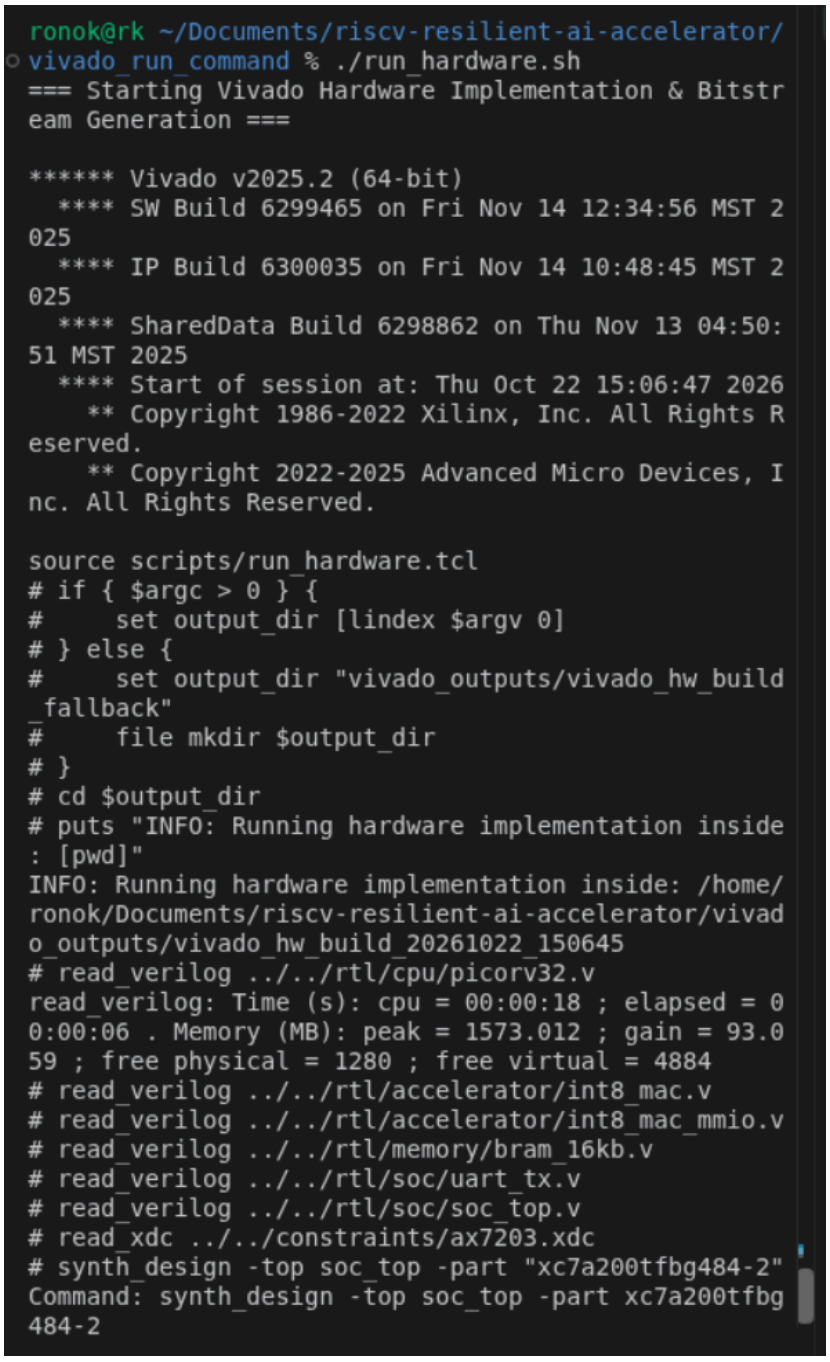
2. Synthesis & Hardware Implementation

- **Synthesis & Routing:** Logic synthesis, optimization, placement, and routing successfully executed for the Artix-7 (xc7a200tfbg484-2) target.

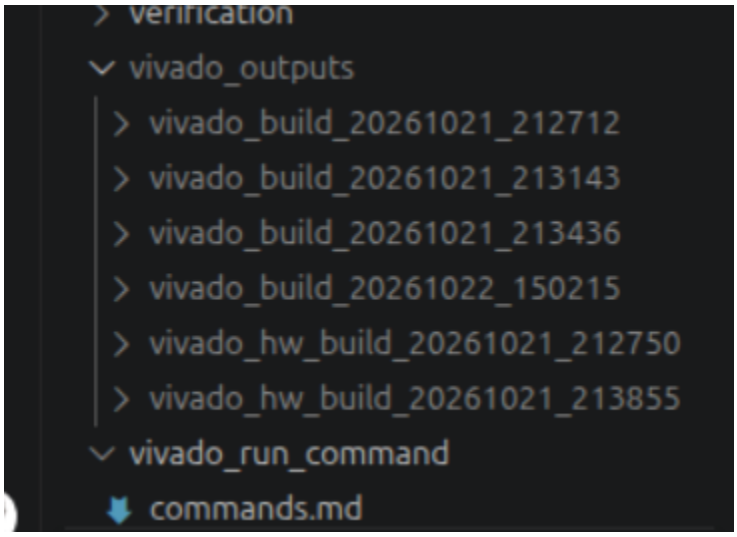
Output:

```
ronok@rk ~/Documents/riscv-resilient-ai-accelerator/
vivado_run_command % ./run_synth.sh
#     set output_dir [lindex $argv 0]
# } else {
#     set output_dir "vivado_outputs/vivado_build_fa
llback"
#     file mkdir $output_dir
# }
# cd $output_dir
# puts "INFO: Running synthesis inside: [pwd]"
INFO: Running synthesis inside: /home/ronok/Document
s/riscv-resilient-ai-accelerator/vivado_outputs/viva
do_build_20261022_150215
# read_verilog ../../rtl/cpu/picorv32.v
# read_verilog ../../rtl/accelerator/int8_mac.v
# read_verilog ../../rtl/accelerator/int8_mac_mmio.v
# read_verilog ../../rtl/memory/bram_16kb.v
# read_verilog ../../rtl/soc/uart_tx.v
# read_verilog ../../rtl/soc/soc_top.v
# read_xdc ../../constraints/ax7203.xdc
# synth_design -top soc_top -part "xc7a200tfbg484-2"
Command: synth_design -top soc_top -part xc7a200tfbg
484-2
Starting synth_design
Attempting to get a license for feature 'Synthesis'
and/or device 'xc7a200t'
INFO: [Common 17-349] Got license for feature 'Synth
esis' and/or device 'xc7a200t'
INFO: [Device 21-403] Loading part xc7a200tfbg484-2
INFO: [Synth 8-7079] Multithreading enabled for synt
h_design using a maximum of 7 processes.
INFO: [Synth 8-7078] Launching helper process for sp
awning children vivado processes
INFO: [Synth 8-7075] Helper process launched with PI
D 147686
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Starting RTL Elaboration : Time (s): cpu = 00:00:05
; elapsed = 00:00:05 . Memory (MB): peak = 2230.949
; gain = 481.188 ; free physical = 133 ; free virtua
l = 4168
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INFO: [Synth 8-6157] synthesizing module 'soc_top' [
/home/ronok/Documents/riscv-resilient-ai-accelerator
```

- **Bitstream Generation:** Successfully compiled and generated the hardware configuration file: **soc_top.bit**.

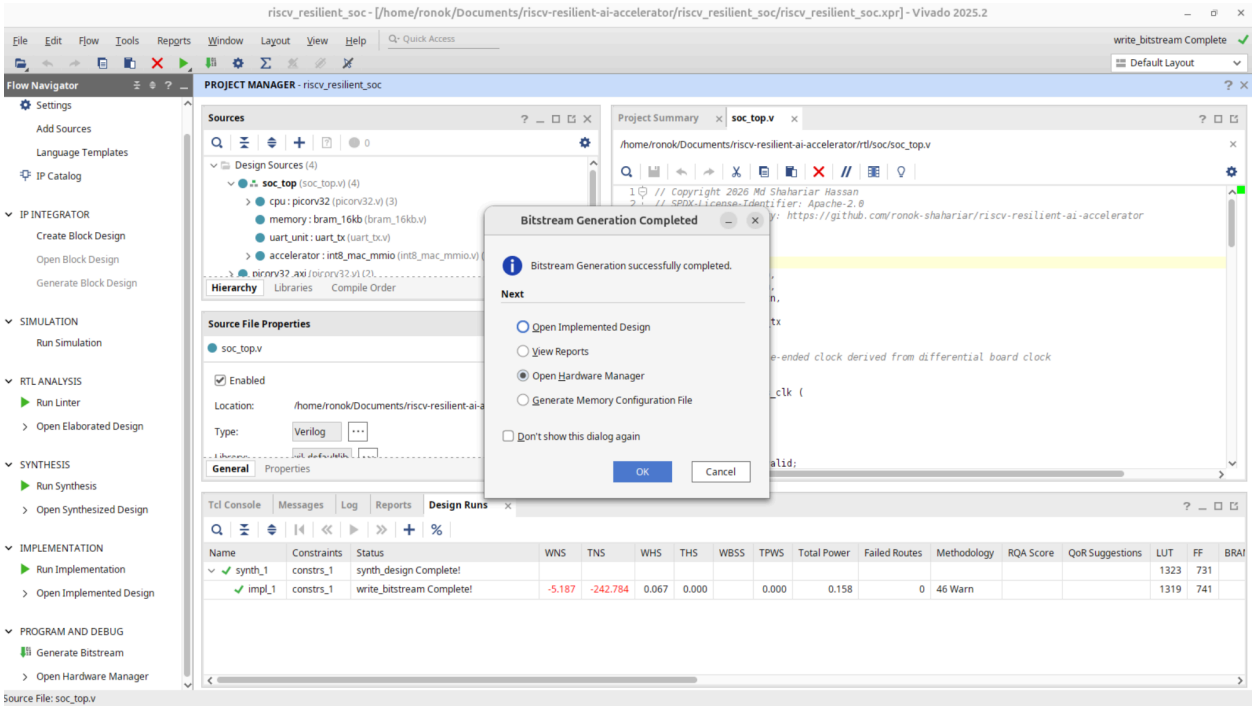


Output Logs:

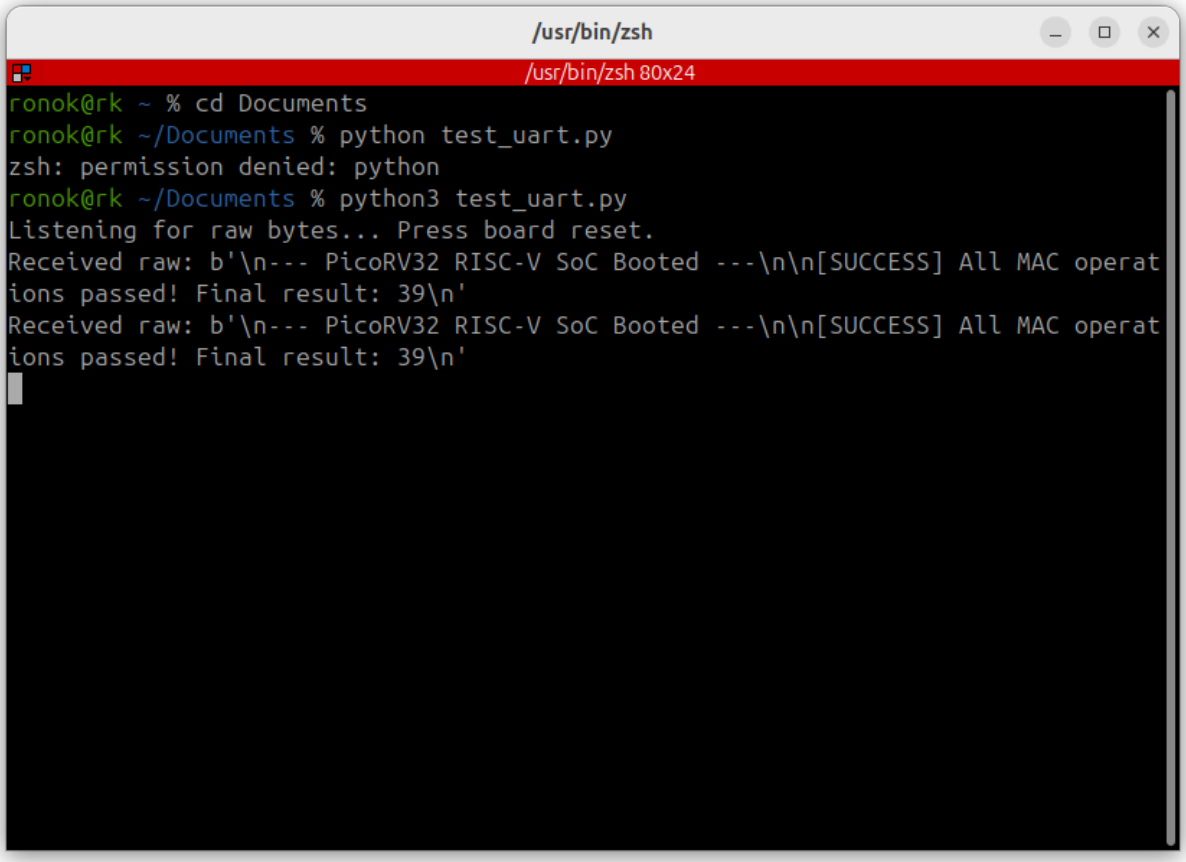


3. Hardware-in-the-Loop & UART Verification

- **FPGA Programming:** Programmed the ALINX AX7203 board via the Vivado Hardware Manager.



- **Runtime Telemetry:** Captured live UART serial output confirming successful SoC boot and execution:



Final Verdict: The entire pipeline—from RTL simulation and synthesis to physical FPGA implementation and hardware verification—is fully functional and verified successfully.